Barcode Checksum Rescue

Catch the fake barcode before the warehouse ships the wrong part.

Win condition: Most bad codes detected with few false alarms.

THE SCIENCE YOU ARE USING

Check digits catch errors. A barcode includes an extra check digit computed from the others by a fixed rule. If a digit is mistyped or smudged, the rule no longer matches and the scanner rejects the code. This is how real product codes catch errors automatically.

Detect without false alarms. A good detector flags the truly bad codes while passing the good ones. You race to find corrupted codes, keep false alarms low, and explain the check-digit rule you used.

HOW TO BUILD AND RUN IT

01 **Learn the rule.** Study the check-digit rule: how the last digit is computed from the rest.

02 **Verify a good code.** Apply the rule to a valid code and confirm it checks out.

03 **Hunt corrupted codes.** Scan a deck of cards and flag the ones that fail the rule.

04 **Avoid false alarms.** Double-check before flagging; wrongly rejecting a good code costs points.

05 **Explain the rule.** Describe how the check digit catches a single mistyped digit.

Safety: Normal classroom management. Allergy: None.

MATERIALS

Printed barcode cards	set
Dry erase boards	per team
Timers	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________

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The ONE variable we are testing: ______________________

Baseline result: ______________________

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct detections	45
Low false alarms	20
Rule explanation	20
Teamwork	15
Total	**100**